EvidenceGraph

Claim–Evidence–Method–Workflow Graphs for Verifiable Scientific Literature Navigation

Complete concept, critical prior-art review, methods, experimentation, public tool architecture, manuscript plan, journal tiers, resources, risks, and publication gates

Research planning edition | 25 October 2026 | Proposed work; no experiments claimed

Executive decision

Build an openly usable evidence-navigation tool focused initially on neuroimaging literature. Its scientific contribution must be tested against paper-chat/RAG, scientific claim-verification, citation-context, knowledge-graph, and executable-paper approaches. The defensible research question is whether explicit claim-level evidence provenance linked to methodological and executable artifacts improves attribution, calibrated uncertainty, contradiction analysis, and method transferability. A software demo or knowledge graph alone is not sufficient proof of novelty.

At-a-glance research strategy

Dimension	Plan
Primary domain	Hippocampal segmentation/asymmetry in epilepsy; optional connectomics transfer
Public product	EvidenceGraph: literature explorer, source-linked claims, methods, provenance and workflow links
Primary evaluation	Expert-grounded claim/evidence tasks; held-out-paper QA; provenance and contradiction evaluation
Proposed main outcome	Verified claim-evidence attribution accuracy (pre-register precise operational definition)
Initial publication path	Methods article if broad performance improvements; software/resource article if narrower impact
Stretch journals	Nature Machine Intelligence; Nature Computational Science
Strong alternatives	Nature Communications; PLOS Computational Biology; potentially biomedical informatics journals
Research status	Protocol, not completed literature systematic review or validated system

1. Scientific motivation and specific claims

Scientific publications communicate findings primarily in prose, while supporting evidence is distributed across paragraphs, figures, supplements, protocols, analysis scripts, datasets and independent citations. A

conventional retrieval assistant can cite an article yet still misstate what the authors actually measured, overstate external validation, confuse supporting versus contradictory findings, or imply that code reproduces a result without verifying it.

EvidenceGraph proposes a typed, inspectable representation that distinguishes reported claims from evidence statements, statistical estimates, population conditions, methods, code, computational runs, and uncertainty. Its reasoning layer is an optional user interface—not the sole source of truth. Claims can remain disputed or unevaluated.

1.1 Proposed contributions

- C1. A schema for source-anchored scientific claim and evidence provenance, including study scope, polarity, evidence type, extraction status, confidence, and missingness.
- C2. A method for linking claims to documented computational methods and executable research objects without asserting reproduction when code or data are unavailable.
- C3. A quality-control workflow combining constrained extraction, cross-document linking, deterministic provenance checks and expert adjudication.
- C4. A benchmark and ablation analysis testing whether graph relationships improve trustworthy research answers beyond strong retrieval and graph baselines.
- C5. A usable public evidence navigator for a selected neuroimaging corpus, with openly licensed annotations and reproducible evaluation.

1.2 Hypotheses (prospective—not observed)

- H1: Source-anchored claim graphs improve evidence attribution and reduce unsupported cross-paper synthesis relative to strong RAG baselines.
- H2: Explicit population/method/context constraints improve correct handling of apparent contradictions and transferability questions.
- H3: Links to executable workflow and provenance artifacts improve assessment of reproducibility status, when artifacts exist.
- H4: Extractor+graph+verification outperforms extractor-only and graph-only configurations on held-out papers; benefits may be task-specific.

1.3 What the work does NOT claim

- That every scientific claim can be decided true or false.
- That a citation or high citation count establishes scientific validity.
- That paper-related code proves the paper is reproducible.
- That automated relation extraction is ground truth.
- That an agent can perform a systematic review without a validated selection, appraisal and reporting protocol.

2. Prior-art landscape and novelty audit

This is a targeted, source-verified narrative review and a seed bibliography, not a PRISMA systematic review. Search comprehensively and update before writing the final manuscript. Distinguish standards, released products, methods papers, evaluation datasets and emerging preprints. The innovation needs to survive comparisons against these established capabilities.

Prior work	Established capability	Non-duplicative direction to test
Paper2Agent (Miao et al., Nature 2026)	Converts papers and code into agent-accessible MCP resources/tools; tests execution against reference outputs	Cross-paper claim provenance, evidence scope and independent validity reasoning rather than paper-agent creation
SciFact (Wadden et al., EMNLP 2020)	Scientific claim support/refute retrieval and rationale evaluation	Link method assumptions, context and executable outputs; assess calibrated multi-paper uncertainty
Evidence Inference	Clinical trial intervention–comparator–outcome evidence extraction	Broader method and provenance relations in neuroimaging; external evidence transfer
Scite smart citations	Citation contexts categorized supporting/contrasting/mentioning	Explicit scientific claim identity, context qualifiers, method equivalence and reproducibility
Open Research Knowledge Graph / scholarly KGs	Structured research comparisons and scholarly assertions	Reproducible artifact links and empirically tested scientific reasoning
RO-Crate / Workflow Run RO-Crate	Machine-readable research objects and workflow execution provenance	Claim-level linkage and reasoning without replacing existing provenance
W3C PROV-O / nanopublications	Foundational provenance and assertion structures	Domain schema and experimentally evaluated graph construction
Europe PMC / OpenAlex	Article retrieval, metadata and scholarly relationships	Expert-validated provenance-aware research navigation

2.1 Key closest competitor: Paper2Agent

Miao et al. (Nature, 16 Sept 2026; DOI 10.1038/s41586-026-11044-y) show paper-to-MCP conversion and execution for computational paper methods, with agents accessing manuscript and supplemental assets. Their paper substantially constrains novelty: publishing “papers become AI agents” is no longer enough. Explicitly compare on claims requiring multiple papers, contradictory evidence, unsupported conclusions, and methods-context distinctions. Do not inaccurately describe Paper2Agent as unable to answer multi-paper questions; it supports connecting multiple paper MCPs.

2.2 Closest evaluation baseline: SciFact

SciFact (Wadden et al., EMNLP 2020; DOI 10.18653/v1/2020.emnlp-main.609) includes approximately 1,400 expert-authored claims with supporting/refuting evidence and rationales. Reuse evaluation conventions when compatible. However, the proposed application requires new gold annotations for claims conditioned on imaging modality, segmentation protocol, cohort and acquisition context.

2.3 Core provenance/research-object standards

Use W3C PROV-O to record agents, entities, derivations and activities; RO-Crate to package datasets/code/metadata; Workflow Run RO-Crate for execution details. Avoid creating competing duplicate representations. Where possible, use typed mappings to existing ontologies and persistent paper/data/software identifiers. Record exact version and provenance even when no runnable workflow is available.

2.4 Literature search strategy and completeness gate

- Databases: PubMed, Europe PMC, Semantic Scholar/OpenAlex metadata, ACL Anthology, ACM/IEEE where available, Crossref, Google Scholar hand-check.
- Keyword clusters: scientific claim verification; scientific evidence graphs; citation intent; research knowledge graphs; scientific RAG; evidence attribution; executable papers; reproducibility; workflow provenance; neuroimaging biomarkers.
- Snowball backwards/forwards from Paper2Agent, SciFact, Evidence Inference, RO-Crate and ORKG.
- Use a screening matrix: year; domain; dataset; claims type; evidence representation; executable links; baseline; metrics; gaps; reproducibility; licensing.
- Before submission: rerun searches, record exact dates, complete full bibliographic metadata/DOIs, verify journal article versions, and add relevant contrary results.

3. Formal task and data model

Define a typed heterogeneous graph G=(V,E,A), where V includes Paper, Claim, EvidenceSpan, Study, Cohort, ImagingProtocol, Method, Dataset, CodeArtifact, Workflow, WorkflowRun, QuantitativeResult, and Limitation. Edges have explicit semantics, provenance and uncertainty, rather than generic “related_to” links.

Node/edge	Required fields	Validation rule
Claim	claim_id, normalized text, source span, epistemic modality, population scope	Never mark published statement as verified without independent adjudication
EvidenceSpan	article DOI/PMCID, paragraph/offset, figure/table pointer, license	Offset and source must resolve to unchanged source version
Supports/Contradicts/Qualifies	claim, evidence, relation type, reviewer status	Relation is contextual, not inherent citation polarity
Method	pipeline, parameter, comparator, measurement definition	No equivalence across methods absent documented mapping
Workflow/Run	code ID, container digest, inputs, run status, outputs	Execution not asserted unless tested and logged
Population/Acquisition	cohort size, disease, scanner, protocol, inclusion criteria	Unknown fields explicitly retained
Uncertainty	reported CI/effect; evidence quality; extraction confidence	Do not conflate extraction confidence with empirical uncertainty

3.1 Example structured record (illustrative only)

claim_id: C001 | asserted_by: DOI placeholder | statement: “Hippocampal asymmetry is associated with pathology-defined HS in the studied cohort” | population: surgical epilepsy cohort | method: MRI segmentation + pathology comparator | evidence_span: exact source offsets | relation: reported_support | effect_estimate: sourced or unknown | limitations: single-site or unknown | verification: pending_expert | workflow_run: unavailable.

3.2 Epistemic states

- Reported: manuscript explicitly asserts the claim.
- Evidence linked: one or more source spans are mapped to the assertion.
- Expert adjudicated: qualified reviewers assess relation and contextual correctness.
- Computationally reproduced: a documented run achieved specified reference comparison criteria.
- Independently replicated: separate study or data support a sufficiently comparable result.
- Contested: contrary evidence exists under context determined by expert assessment.
- Unknown/not assessed: insufficient evidence or missing artifacts.

These states are not a linear “truth score”; they can coexist. For example, a result might be computationally reproduced but have uncertain external validity.

3.3 Graph consistency constraints

- Every factual answer must carry exact provenance at the sentence or claim level.
- Never infer support from citation existence alone.
- Prevent relation labels generated without extractable evidence passages.
- Never equate different patient cohorts or image protocols solely by a shared clinical keyword.
- Treat missing code as unknown reproducibility, not failure to reproduce.
- Carry publication/version dates and retraction/correction flags where reliably available.
- Enforce permissions and article license boundaries on text storage and public dissemination.

4. EvidenceGraph public tool specification

Deliver a practical public product alongside the methodology. First release is a research literature explorer, not autonomous clinical decision support or an unsupervised systematic-review substitute.

4.1 User personas and core journeys

- Graduate student: identify the exact studies behind a biomarker claim and inspect methods.
- Neuroimaging engineer: compare segmentation/registration requirements and discover available code.
- PI/systematic reviewer: identify conflicting results conditioned on population, scanner and reference standard.
- Computational scientist: follow a study to available workflow, data, provenance and reproducing command.

4.2 MVP features

- Import DOI/PMCID or curated permitted full text; resolve metadata and license.
- Paper reader with linked highlighted source passages.
- Claim cards showing exact reported wording, normalized paraphrase, source and qualifiers.
- Evidence graph visualization with filters for population, method, strength, direction and status.
- Method comparison matrix across papers.
- Citation export and machine-readable evidence record export (JSON-LD).
- Manual annotation review and correction history.
- Clear “unknown” and “not independently checked” labels on every answer.

4.3 Next-stage features

- Automatic evidence question decomposition with constrained retrieval.
- Cross-paper contradiction and methodological transferability assistant.
- Connection to SciCap registered capabilities and SciGuard validation outcomes.
- Workflow execution/provenance links through RO-Crate; permission-aware run/reproduction requests.
- Collaborative expert curation with adjudication and versioned releases.

4.4 Reference architecture

INGEST: Europe PMC/PMC permitted full text + Crossref/OpenAlex metadata → normalize paragraphs/sections/citations with identifiers → extraction jobs with quoted spans → schema validation → expert review queue → graph storage and full-text index → retrieval API → citation-constrained synthesis UI. Separate services handle executable object resolution and provenance. A graph database (Neo4j) or RDF triplestore is optional initially; JSON-LD + PostgreSQL may be more maintainable for an MVP.

Subsystem	Suggested implementation	Design justification
Corpus and metadata	Python, Europe PMC APIs, DOI/PMCID resolver	Licensed ingestion and persistent IDs
Extraction	Structured LLM or biomedical NLP + rules	Span anchoring and schema-constrained output
Graph storage	PostgreSQL/JSONB + graph queries; later Neo4j/RDF	Avoid premature graph complexity
Vector/text retrieval	PostgreSQL full text or	Strong non-graph baseline and

	OpenSearch; optional vectors	production search
Evidence API	FastAPI/Pydantic, versioned JSON-LD	Machine-readable assertions
Web UI	React; citation/highlight and graph views	Accessible user inspection
Provenance links	RO-Crate, PROV, Git commit and container digests	Do not duplicate workflow metadata
Evaluation	Python scorer, gold annotations, fixed splits	Reproducibility and blinded scoring

4.5 Public release and security

Open code and redistributable annotations; do not republish restricted article full text. Respect PMC automated retrieval mechanisms and per-article licenses. Make public datasets the default. External user data or PHI must not be uploaded by default. Enable role-based write access, annotation audit trails, versioned APIs, opt-in analytics and explicit correction processes. Distinguish machine output from expert-reviewed evidence.

5. Corpus strategy and acquisition

Start narrowly with hippocampal asymmetry, hippocampal sclerosis and MRI segmentation in epilepsy, then expand into structural connectomics as a held-out transfer domain. This leverages domain knowledge without making the benchmark dependent on inaccessible institutional data.

Phase	Target corpus / annotation	Source and eligibility
Pilot	20–30 papers; 100–200 claim-evidence edges (planning values)	Open-access neuroimaging studies; expert-screened; licensing verified
Main corpus	80–150 papers; 500–1,000 adjudicated relations (planning values)	Mix of positive, null and conflicting findings; representative methods
External transfer	30–50 papers from distinct imaging or biomedical domain	Held out from schema tuning and annotation authoring
Executable subset	10–20 papers with code/data and runnable references (conditional)	Only where legal and technically reproducible

These target sizes are planning assumptions, not statistical sample-size justifications. Before full annotation, pilot inter-rater reliability and define the minimum useful confidence-interval width for the primary accuracy endpoint. Use sample-size simulation/power analysis for paired benchmark conditions.

5.1 Data sources

- Europe PMC: article metadata and eligible open access content; follow API and licensing requirements.
- PMC Open Access subset: obtain allowed full text through authorized services; check article license individually.
- PubMed: metadata and biomedical indexing for coverage.
- Crossref/OpenAlex: DOI resolution, work metadata and scholarly links; not substitutes for licensed full text.
- Public GitHub/Zenodo/OSF repositories: source-linked code and release artifacts, with individual license review.
- SciFact and Evidence Inference: use for supplementary transfer or baseline evaluation if their usage terms permit.

5.2 Selection and annotation protocol

- Prespecify scope and query strings; document screening decisions and excluded reasons.
- Stratify papers by positive/null findings, studies with and without executable code, publication year, sample size and modality.
- Define annotation handbook with worked cases: hypothesis vs finding, association vs causal claim, subgroup qualifier, apparent contradiction, and missing evidence.
- Two independent annotators for a substantial subset; adjudicate disagreements before gold release; report agreement by edge type.
- Use sentence/paragraph-level provenance; record source version and offsets; expert review must inspect the source rather than LLM summaries only.

- Keep paper-level held-out sets to prevent paragraph overlap across train/dev/test; time split as secondary test.

5.3 Ethics and data governance

Do not ingest restricted clinical notes or patient images for the initial corpus. Literature annotation may not constitute human-subject research, but study of tool users, interviews, identifiable usage analytics, or institution-linked resources may require an institutional determination. Establish a documented license matrix and user contribution agreement before accepting external annotations.

6. Experimental design and statistical analysis

6.1 Research questions and outcomes

Question	Primary measure	Secondary measures
RQ1: attribution	Correct claim–source–span triple under expert gold criteria	Unsupported statement rate; span localization F1
RQ2: context	Correct interpretation of population/method qualifiers	Context omission; contradiction error; abstention
RQ3: executable evidence	Accurate identification of reproducibility status	False claims of reproduction; method link precision
RQ4: system value	Paired task quality vs strongest baseline	Cost, latency, expert review burden, calibration

6.2 Fair baseline matrix

System	Available information	Purpose
BM25/full-text search	Same indexed permitted corpus	Retrieval-only control
Dense/vector RAG	Same corpus, passages, model and budget	Modern retrieval control
Hybrid retrieval + LLM	Lexical + vector, citations	Strong non-graph production baseline
Knowledge graph RAG	General graph structure and retrieval	Controls for adding a graph
SciFact-style verifier	Claim-evidence classification	Controls for support/refute behavior
Paper2Agent-style paper access where feasible	Paper/tool artifacts, same reachable source set	Controls for executable-paper capabilities
EvidenceGraph full	Anchored claims + scoped evidence + methods/provenance	Tests proposed innovation

Avoid an unfair baseline by withholding full text, code or context from competing systems. If Paper2Agent is evaluated in only a subset due to deployment constraints, state the subset and do not imply full comparable replication.

6.3 Experiments

1. E1 Extraction quality: compare structured extraction to manual annotation; span precision/recall, normalized claim accuracy, qualifier recall, calibration.
2. E2 Evidence QA: fixed questions requiring one or several papers; blinded expert grading for correct claims, evidence, uncertainty and omissions.
3. E3 Contradiction: paired studies with conflicting or context-dependent results; distinguish genuine disagreement from population/protocol differences.

4. E4 Transferability: ask whether an established method applies to a different MRI protocol/cohort; require appropriate abstention when not supported.
5. E5 Reproducibility state: identify code/data availability, run status and agreement with published outputs; execute only documented subset with permissions.
6. E6 Ablation: drop evidence spans, remove study-context nodes, remove method links, remove execution provenance, disable expert-reviewed edges, and test zero vs few-shot extraction.
7. E7 Generalization: held-out publication years, laboratories, and a second domain; test robustness to noisy text/OCR and missing code.
8. E8 Human study (optional): compare researcher time-to-evidence, citation correction effort and confidence calibration under ethics-approved protocol.

6.4 Statistical plan

Pre-register one primary outcome, such as the percentage of answered benchmark items whose central factual claims have correct evidence spans and appropriate scope. Perform paired comparisons on identical items; estimate clustered bootstrap confidence intervals resampling at the paper level, not only at the question level. For binary paired outcomes, consider McNemar or mixed-effects logistic models as appropriate. Report multiple-testing correction for secondary endpoints, exact denominators, effect sizes, interval estimates, performance by question type and abstention trade-offs.

Score unsupported answers separately from appropriate abstention. Use a blinded rubric (claim accuracy, source exactness, context qualifier, uncertainty calibration, unsupported generalization) and measure inter-rater agreement. Report model/version, sampling settings, prompts, number of repeats, hardware, token usage, and per-query cost.

6.5 Failure analysis

- Citation exists but does not support the generated statement.
- Correct evidence passage, incorrect population or comparator.
- Method similar in name but not equivalent in implementation.
- Null result paraphrased as evidence of no effect.
- Missing code treated as failed reproducibility.
- AI extraction leaks an inference as a verbatim reported result.
- A contradictory paper is misclassified because timepoint/outcome definitions differ.
- Graph retrieval selects a topologically near but epistemically unrelated claim.

7. Manuscript package

7.1 Proposed article title variants

- EvidenceGraph: Provenance-Aware Scientific Claim Reasoning Linked to Reproducible Methods
- From Claims to Computation: Source-Grounded Evidence Graphs for Scientific Literature Navigation
- Evaluating Evidence-Method Graphs for Reliable Cross-Paper Scientific Reasoning

7.2 Structured abstract scaffold (no fabricated results)

BACKGROUND: Scientific findings are distributed across claims, supporting passages, methods and artifacts, hindering context-sensitive reuse. METHODS: We propose a source-anchored claim-evidence-method graph and a public evidence navigator; we evaluate it against lexical, RAG, graph-RAG and claim-verification baselines on expert-curated neuroimaging questions. RESULTS: [Insert preregistered endpoint estimates, uncertainty and subgroup outcomes after experiments]. CONCLUSIONS: [State only demonstrated improvements and their limitations].

7.3 Full manuscript section map

9. Introduction: reproducibility challenge, scientific attribution, narrow gap beyond paper agents and fact verification.
10. Related work: claim verification, evidence extraction, scholarly KGs, RAG, paper agents, RO-Crate/provenance and domain-specific literature synthesis.
11. Methods—Corpus: sources, inclusion/exclusion, licenses, split strategy, expert annotation, agreement.
12. Methods—Formalism: graph ontology, uncertainty states, anchored edges, method and run representation.
13. Methods—System: extraction, validation, indexing, graph retrieval, agent synthesis, user interface.
14. Methods—Evaluation: tasks, baselines, fairness controls, primary endpoint, statistical protocol.
15. Results: annotation reliability; claim extraction; primary QA; contradiction/context; ablation; transfer; computational reproducibility subset.
16. Discussion: contribution boundaries, impact on research navigation, missing evidence, generalization, computational burden, user controls.
17. Availability/ethics: code, gold data, prompts, artifacts, privacy, licenses, conflicts and AI disclosure.

7.4 Figures and tables

Item	What to present
Figure 1	Problem and source–claim–evidence–method–workflow graph architecture
Figure 2	End-to-end tool architecture and article-to-graph pipeline
Figure 3	Primary attribution results with confidence intervals and baselines
Figure 4	Contradiction/context task strata and abstention curves
Figure 5	Ablation of graph components and expert validation

Figure 6	Executable method/provenance case study with honest failure status
Table 1	Literature competitor matrix and functionality overlap
Table 2	Corpus composition, licenses and expert agreement
Table 3	Baseline configurations, costs and statistical results
Supplement	Annotation manual, full prompts, task schema, examples, reproducibility checks

7.5 Reviewer objections and required answer

- “This is graph-RAG with extra metadata”: quantify contribution of provenance, method and evidence semantics through ablation.
- “Paper2Agent already does this”: evaluate a distinct claim-level task and include paper-agent comparator where feasible.
- “Graph edges are hallucinated”: require anchored spans, audited extraction and independent expert adjudication.
- “Only epilepsy literature”: hold out a separate domain and acknowledge generalization limits.
- “No scientific discovery”: report tangible methodological or evidence-interpretation improvement; avoid unvalidated discovery claims.
- “The gold set is biased”: preregister selection, publish handbook, include external reviewers and report difficult negative cases.
- “User study is anecdotal”: use controlled within/between-subject design and validated endpoints if such claims are made.

8. Journal tiers and explicit publication gates

Journal tiers below are strategic fit/ambition, not guarantees, quartile assignments or acceptance probabilities. Evaluate article type and author instructions again before submission.

Journal / tier	Required claim and evidence	Decision gate
Nature Machine Intelligence — A+ stretch	Novel broadly relevant AI reasoning/representation result, major robust baselines, multi-domain expert validation and generalizable insights beyond a literature UI	Do not submit with neuroimaging-only demo or minor RAG gains
Nature Computational Science — A+ stretch	General computational framework linking evidence and executable research; independent domains, substantive methodological improvement, rigorous reproducibility	Need cross-domain, consequential scientific computing results
Nature Communications — A ambitious	Important advance for specialist audiences, conclusive and well-controlled empirical evidence, clear novel contribution	Sustained improvements with convincing mechanistic insight
PLOS Computational Biology — A domain-fit	Outstanding method/software enabling biological insight; broad adoption potential, real biological application, complete open code/data	Strong biomedical methods contribution and reproducible real-data validation
Journal of Biomedical Informatics / JAMIA — strong specialty	Biomedical knowledge/evidence methods and trustworthy evaluation with clinical/informatics relevance	Focus on clinical-context evidence questions and safety boundaries
GigaScience / SoftwareX — practical specialist	Open and reusable data/workflow/software with substantial technical validation and documentation	If method novelty insufficient but public tool is robust

8.1 Thresholds to set before first submission

- Define minimum effect size worth publishing (e.g., percentage-point gain in fully supported answer rate) before seeing main outcomes; choose based on pilot variance and usefulness, not an arbitrary universal cutoff.
- For elite journals, require strong cross-domain and cross-model demonstration, independent expert replication, compelling ablation and major methods advance.
- For methods specialty journals, require reproducible baseline superiority, realistic biomedical use cases, clear limitations and public framework implementation.
- For software-oriented journals, prioritize installation, tests, robust community documentation, licensing, release versioning and demonstrated research use.

- Avoid journal shopping through unsupported claims; select venue once main contributions and paper type are evident.

8.2 Strategic recommendation

Develop EvidenceGraph initially as a public, expertly curated neuroimaging tool and evaluate claim-level evidence attribution. If results demonstrate broad AI methodology, expand to second/third fields for Nature Machine Intelligence or Nature Computational Science. If benefits are domain-specific, target PLOS Computational Biology or biomedical informatics outlets. Preserve a distinct software/resource paper only when it makes an independent contribution and does not duplicate the methods paper.

9. Implementation milestones and resources

Stage	Deliverables	Exit gate
0. Governance + review	Licenses, institutional ownership, systematic search protocol, annotation rubric	Permitted corpus and approved research scope
1. Pilot corpus (weeks 1–5)	20–30 papers; paragraph IDs; initial extraction; expert cases	Resolvable anchors and preliminary agreement
2. Public MVP (weeks 6–10)	Source viewer, claim graph, filters, JSON export, corrections	Independent user can reproduce import and inspect evidence
3. Benchmark (weeks 11–16)	Fixed task splits, rubrics, blind adjudication, baseline harness	Baseline runs and data governance documented
4. Main study (weeks 17–25)	Experiments E1–E7, statistical analysis and failures	Primary endpoint and uncertainty complete
5. Cross-domain and manuscript	External corpus, figures, repo release, paper	Novelty and journal fit reassessed after evidence

These are tentative work stages, not promised durations. Corpus rights, annotation capacity and executable-paper availability may dominate calendar time.

9.1 Team

- Research engineering lead: ingestion, graph/API, tool, benchmark and reproducible runs.
- Neuroimaging methods expert: cohort/method terminology and gold labels.
- Biomedical NLP/knowledge representation collaborator: extraction, ontology and evaluation rigor.
- Information specialist/librarian: literature search and license/source guidance.
- Biostatistician/evaluation methodologist: sample-size plan and clustered inference.
- Independent annotators: agreement studies; conflicts resolved by qualified adjudicator.

9.2 Budget and computational management

Design an initial low-cost pilot using only 20–30 documents, batched local embeddings, cached retrieval and a constrained LLM call budget. Track annotator hours, extraction attempts, token use and inference cost. Full-text processing has copyright constraints irrespective of technical access. A hosted graph UI need not run heavy model inference; offer local/demo options and rate limits.

9.3 Relationship to Papers 1–3 and 5

EvidenceGraph provides the literature/evidence layer. SciCap can index method capabilities referenced in papers; SciGuard can validate scientific workflow prerequisites; NeuroAgentBench can host independent agent-evaluation suites; NeuroInsight can execute accessible workflows under authorization. These integrations should be optional, well specified, and not required to demonstrate the Paper 4 core claim. Avoid recycling identical benchmark tasks or results across manuscripts without transparent cross-reference.

10. Research reproducibility and scientific governance

- Freeze exact article corpora with DOI/PMCID, retrieval date, source version and license matrix.
- Release permitted annotations, schemas, IDs, splits, prompts, software versions, evaluation code and exact experimental settings.
- Maintain public and restricted-data manifests separately.
- Maintain provenance of human corrections and model generation; no anonymous silent graph overwrites.
- Document researcher conflicts, use of automated AI assistance, model limitations and prompts.
- Offer correction/retraction updates for ingested papers where sources permit reliable tracking.
- Pre-register evaluation protocol where feasible; distinguish exploratory and confirmatory results.
- Use a manuscript checklist reviewed by domain and methods collaborators before submission.

11. Completion checklist

- Novelty matrix updated with new work at submission date.
- Full-text rights, provenance identifiers and extraction source offsets verified.
- Schema defined and released with clear semantic relation guidelines.
- Gold annotations independently reviewed and inter-rater statistics reported.
- Baselines strong, fair and computationally reproducible.
- Primary endpoint specified and estimated with appropriate uncertainty.
- Ablations verify exact reasons for observed improvements.
- External generalization evaluated (or scope honestly constrained).
- Executable workflows tested only where appropriate; all untested claims clearly marked.
- Public website/package installs and works for external users.
- Public replication artifacts accessible through archived release with persistent identifiers.
- Manuscript, supplementary methods, data/code availability, author approvals, and journal reporting completed.

12. Verified reference seed bibliography

References below are a curated starting set, not a complete formal systematic review. URLs are provided to enable full-text inspection and accurate reference-manager import; confirm publication details and access conditions before submission.

[1] Miao J, Davis JR, Zhang Y, et al. (2026). Reimagining research papers as interactive and reliable AI agents. Nature. DOI:10.1038/s41586-026-11044-y
https://doi.org/10.1038/s41586-026-11044-y

[2] Wadden D, Lin S, Lo K, et al. (2020). Fact or Fiction: Verifying Scientific Claims. EMNLP. DOI:10.18653/v1/2020.emnlp-main.609
https://aclanthology.org/2020.emnlp-main.609/

[3] SciFact dataset and evaluation repository (Allen Institute for AI)
https://github.com/allenai/scifact

[4] Evidence Inference dataset and annotation download
https://evidence-inference.ebm-nlp.com/download/

[5] Research Object Crate (RO-Crate) Specification v1.3
https://www.researchobject.org/ro-crate/specification

[6] Leo S, Crusoe MR, Rodríguez-Navas L, et al. (2024). Recording provenance of workflow runs with RO-Crate. PLOS ONE 19(9):e0309210.
https://doi.org/10.1371/journal.pone.0309210

[7] Workflow Run RO-Crate documentation and profiles
https://www.researchobject.org/workflow-run-crate/

[8] W3C PROV-O: The PROV Ontology
https://www.w3.org/TR/prov-o/

[9] Europe PMC developer services and APIs
https://europepmc.org/developers

[10] PMC Article Datasets and text mining usage restrictions
https://pmc.ncbi.nlm.nih.gov/tools/textmining/

[11] Nature Machine Intelligence: aims and scope
https://www.nature.com/natmachintell/aims

[12] Nature Computational Science: aims and scope
https://www.nature.com/natcomputsci/aims

[13] Nature Communications: aims and scope
https://www.nature.com/ncomms/aims

[14] PLOS Computational Biology: journal information and article type criteria
https://journals.plos.org/ploscompbiol/s/journal-information

[15] PLOS Computational Biology: submission guidelines
https://journals.plos.org/ploscompbiol/s/submission-guidelines

[16] ORKG: Open Research Knowledge Graph website (review publications and schema documentation)
https://www.orkg.org/

[17] OpenAlex documentation and scholarly metadata
https://docs.openalex.org/

[18] Crossref REST API documentation
https://www.crossref.org/documentation/retrieve-metadata/rest-api/

[19] FAIR Guiding Principles for scientific data management and stewardship (Wilkinson et al. 2016)
https://doi.org/10.1038/sdata.2016.18

[20] Nanopublication architecture and documentation (verify source-specific scholarly references)
https://nanopub.net/

13. Explicit open questions to resolve in the pilot

- What proportion of papers provide machine-extractable passages, reproducible numerical results, code and accessible datasets?
- Are expert annotators consistent in classifying “supports” vs “qualifies” across different study populations?
- Is evidence-provenance representation improving answers beyond the additional token budget or a larger retrieval context?
- How frequently can contradictions be resolved as scope mismatches versus genuine result disagreement?
- What fraction of published methods can be linked to actual executable research objects without manual reconstruction?
- Which data representation offers the best tradeoff among graph query utility, version management and implementation complexity?
- What meaningful endpoint is best aligned with target journals: claim accuracy, reproducibility assessment, or cross-paper context-aware reasoning?

End of blueprint. This planning document intentionally contains no invented experimental results, confirmed acceptance probabilities, or unverified claims of published novelty.